Name: Nachiket Joyekurun Student ID: S224794365 Task 2.1P

Sending Temperature and Light Data to the Web

GitHub link: https://github.com/NachiketJoy/SIT730/tree/main/Task%202.1P

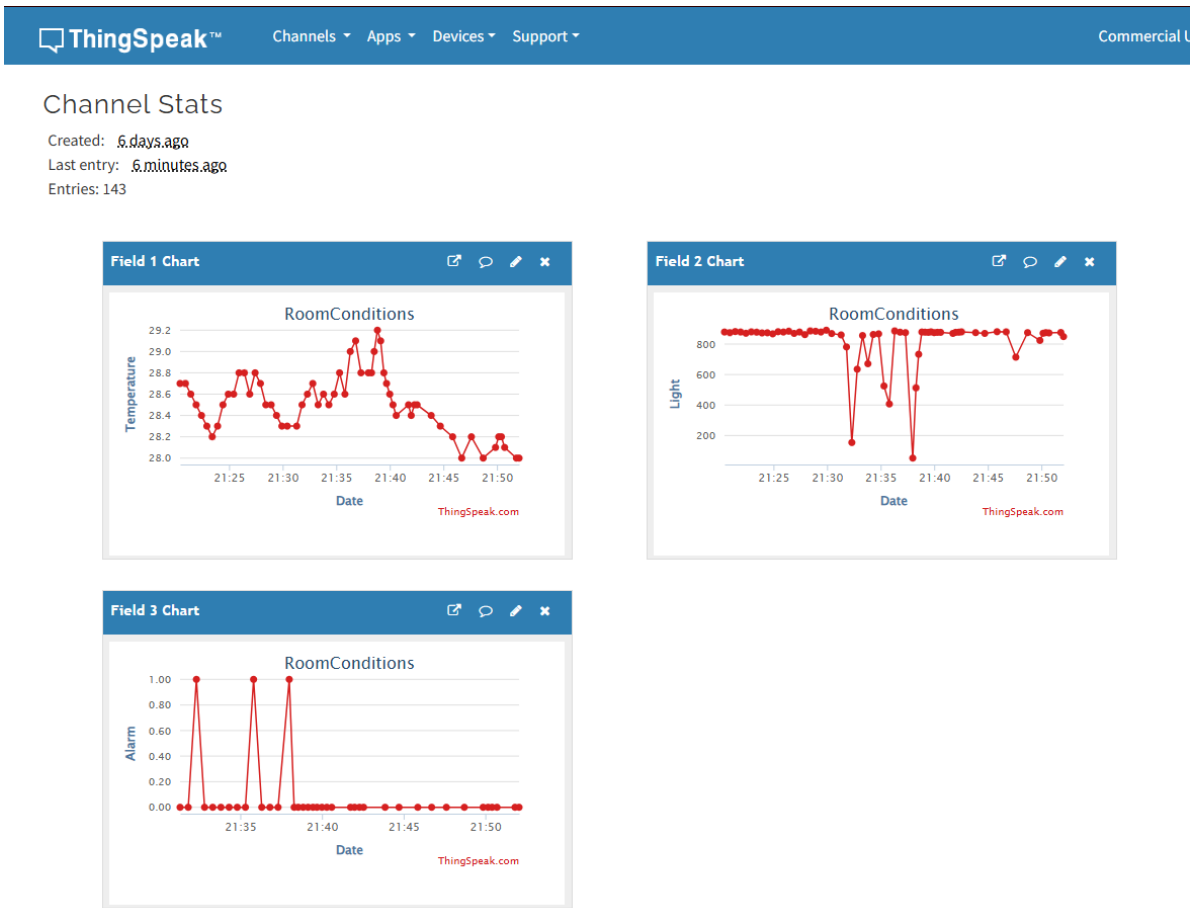
Video link:
https://drive.google.com/file/d/1mB0R6Rnry_4wn4lyJkiijT397b1Um2ZT/view?usp=sharing

This task collects temperature and light data from an Arduino Nano 33 IoT and uploads it to ThingSpeak every 30 seconds. Additionally, an alarm field is added to indicate when the light level falls below a safe range.

Webhooks allow a device or service to push data to a web endpoint automatically whenever new information is available. Instead of waiting for a request, the device pushes the data directly to a web service using a HTTP call as soon as an event occurs. This makes it useful for real-time updates in things like payment alerts, IoT notifications, and automation tools.

In this project, the function sendToThingSpeak() performs this behaviour. It sets the sensor values into ThingSpeak fields and sends them using writeFields(), which issues an HTTP request to the ThingSpeak API. This matches the webhook mechanism because the Arduino pushes new temperature, light, and alarm data to the web whenever the 30-second interval is reached.

ThingSpeak:



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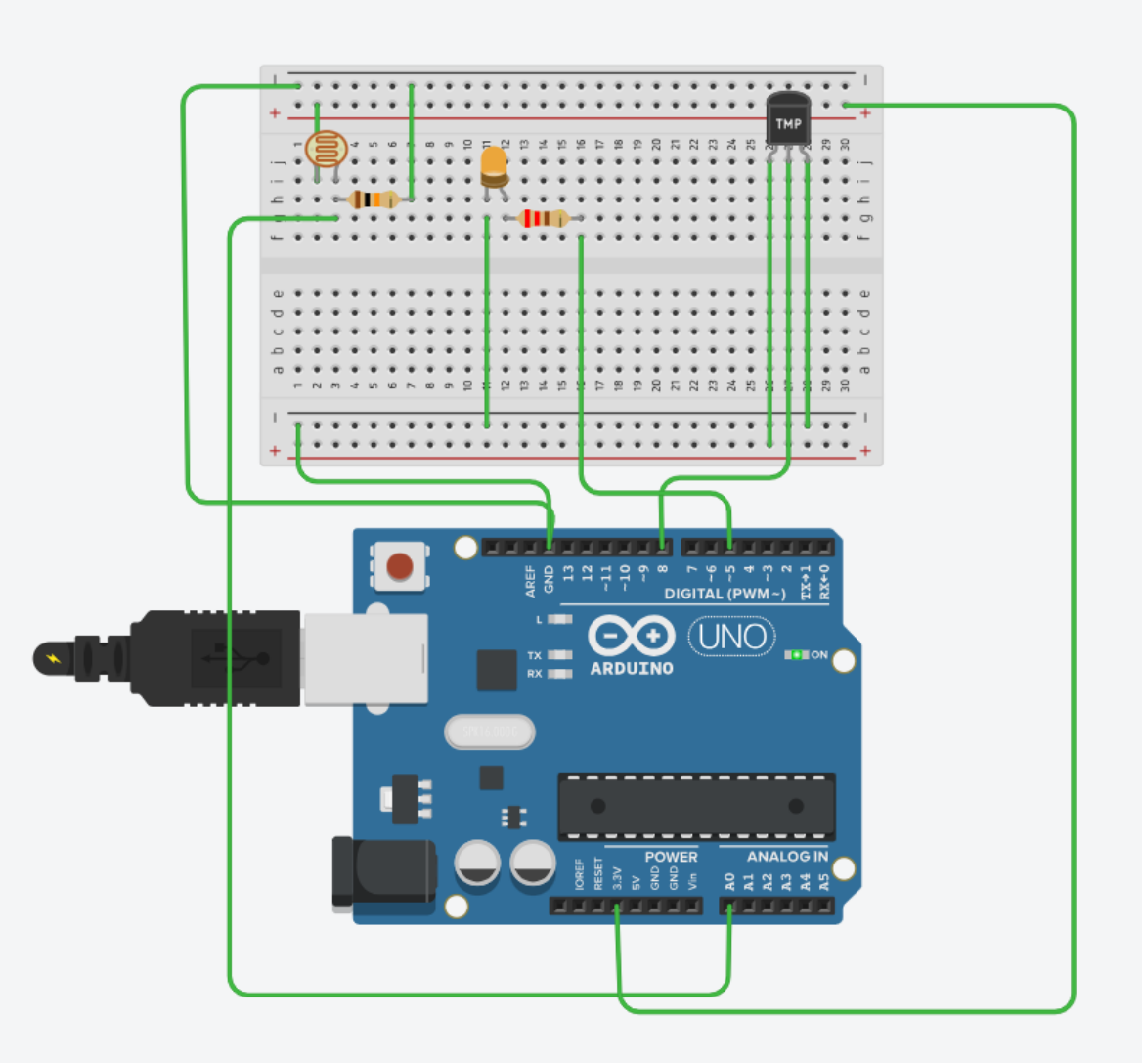
Circuit Diagram: